

Supporting Information for

Machine-learning-assisted screening of interface passivation materials for perovskite solar cells

Chongyang Zhi^{1,2,#}, Suo Wang^{1,2,#}, Shijing Sun³, Can Li^{1,2}, Zhihao Li^{1,2}, Zhi Wan^{1,2},
Hongqiang Wang^{1,2,4,*}, Zhen Li^{1,2,*}, Zhe Liu^{1,2,4,*}

¹*School of Materials Science and Engineering, Northwestern Polytechnical University,
Xi'an, Shaanxi 710072, People's republic of China*

²*State Key Laboratory for Solidification Processing, Northwestern Polytechnical
University, Xi'an, Shaanxi 710072, People's republic of China*

³*Department of Mechanical Engineering, Massachusetts Institute of Technology,
Cambridge, MA 02139, United States*

⁴*Northwestern Polytechnical University -Chongqing Innovation Center, Chongqing
401147, People's republic of China*

[#]*These authors contributed equally*

^{*}*Corresponding authors: hongqiang.wang@nwpu.edu.cn; lizhen@nwpu.edu.cn and
zhe.liu@nwpu.edu.cn*

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1. Experimental details

(a) Chemicals and Materials

Fluorine-doped tin oxide (FTO)-coated and indium-doped tin oxides (ITO)-coated glasses were purchased from Youxuan. Formamidinium iodide (FAI), and methylammonium bromide (MABr) were purchased from Greatcell. PbI_2 and PbBr_2 were purchased from TCI. CsI and all 2D ammonium salts were purchased from Xi'an Polymer Light Technology. Anhydrous dimethyl sulphoxide (99.8%, DMSO), *N,N*-dimethylformamide (99.8%, DMF), chlorobenzene (CB), isopropanol (IPA) and bis (isopropoxide)-bis(acetylacetonate)titanium (IV) was purchased from Sigma-Aldrich. 2-phenylpropan-1-aminium iodide (2-PPAI) and 2-(*p*-tolyl) ethan-1-aminium iodide (2-PTEAI) were purchased from Xi'an E-light New Material. All the chemicals were used as received without further purification.

(b) Fabrication of CsFAMA-based devices

FTO substrates were cleaned by ultra-sonication in 1% Helmanex surfactant aqueous solution, acetone, water, and ethanol in sequence (20 min for each). A compact TiO_2 (c- TiO_2 , thickness ~ 20 nm) blocking layer was deposited onto FTO by spray pyrolysis of 1 mL bis (isopropoxide)-bis(acetylacetonate)titanium (IV) dissolved in 9 mL anhydrous isopropanol (IPA) at 450 °C. After cooling down to room temperature, a mesoporous TiO_2 layer (m- TiO_2 , thickness ~ 150 nm) was deposited by spin-coating at 4000 rpm for 20 s with a TiO_2 colloidal solution (Dyesol 30 NR-D:ethanol=1:7 by weight). The film was dried at 100°C for 10 min followed by sintering at 500 °C for 30 min.

The CsMAFA triple-cation perovskite solar cells were fabricated using a one-step anti-solvent method. The precursor solution was prepared by dissolving FAI (1.19 M), PbI_2 (1.30 M), MABr (0.21 M), PbBr_2 (0.21 M) and CsI (0.07 M) in a mixed solvent of DMF/DMSO (4:1 v/v) with continuously stirring for 2 h. The 3D perovskite film was deposited on the FTO/c- TiO_2 /m- TiO_2 substrate with a two-step spin-coating of 1000 rpm for 10 s and 5000 rpm for 30 s. 120 μL CB was poured onto the substrate 15 s before the end of the second step. The substrate was then annealed at 100°C for 30 min.

2D perovskite layer was formed *in-situ* on top of the 3D perovskite by reacting with ammonium iodide salts IPA solution at 5000 rpm for 30 s, followed by annealing at 100°C for 10 min. The hole-transporting layer (HTL) was deposited on the perovskite film with a receipt of 64 mg spiro-OMeTAD with 24 μ L 4-test-butyl pyridine (tBP), 14 μ L bis(trifluoromethanesulfonyl)imide (LiTFSI, 520 mg/mL in acetonitrile) and 11 μ L tris(2-(1H-pyrazol-1-yl)-4-tert-butylpyridine) cobalt (III) tris(bis(trifluoromethylsulfonyl)imide) (FK209, 376 mg/mL in acetonitrile) in 1 mL of chlorobenzene. The spiro-OMeTAD solution was spin-coated at 4000 rpm for 30 s. Finally, a gold electrode with a thickness of 80 nm was thermally deposited to finish the device. The active area of the device was 0.045 cm².

(c) Fabrication of FAMA-based devices

The SnO₂ ETL was spin-coated on ITO with a SnO₂ colloidal solution (2.67%, diluted with water) at 3,000 rpm for 30 s, and annealed in ambient air at 150 °C for 30 min. The FAMA-based perovskite solar cells were fabricated using a two-step interdiffusion method. Firstly, 1.5 M PbI₂ in DMF: DMSO = 9:1 was spin-coated on SnO₂ at 1,500 rpm for 30 s and then annealed at 70 °C for 1 min. Subsequently, an organic salt solution containing FAI: MAI: MACl (90:6.4:9 mg in 1 mL IPA) was spin-coated on the PbI₂ films at 2,000 rpm for 30 s. The films were then annealed in ambient atmosphere (relative humidity=30%-40%) at 150°C for 15 min. The 2D passivation layer is formed using the same method described above. The spiro-OMeTAD solution was spin-casted on the surface of the perovskite films at 4,000 rpm for 30 s, with a receipt of 72.3 mg spiro-OMeTAD, 28.8 μ L tBP, and 17.5 μ L LiTFSI solution (520 mg/mL in acetonitrile), respectively in 1 mL of chlorobenzene. Lastly, the Ag electrode was thermally evaporated with a thickness of about 100 nm.

(d) Solar cell characterizations

J-V curve characteristics of the PSCs were recorded using a Keithley 2400 source meter under simulated solar radiation of AM1.5G, 100 mW cm⁻² from a Zolix Sirius-SS150A solar simulator. The solar simulator was calibrated using a standard silicon solar cell with a KG5 filter.

2. Machine learning Methods

All the machine-learning scripts were written in Python. The code and the essential data to run the code are available on the GitHub repository (github.com/chrisliu2007/passivation_layer_screening).

(a) Latin Hypercube Sampling (LHS)

LHS is a well-known low-discrepancy pseudo-random sampling method. The LHS method was used to sample new experimental materials within the pre-defined parameter space. LHS equally stratifies the parameter space into smaller hypercubes (e.g., chessboard), and then the sampling procedure follows the rule of exclusivity (e.g., the same rows and columns of the sampled hypercubes are excluded in the subsequent round of sampling. The key advantage of LHS is that this pseudo-random sampling strategy tends to cover more divergent parameter space than random sampling.

The LHS procedure includes the following steps to generate uniform sampling in the halide-amine materials database:

- (1) Divide each range of variables in the parameter space (e.g., the bounds of materials descriptors) into m levels (m equals to the number of samples);
- (2) Randomly select a point within each level of each variable, and samples are picked sequentially so that each level is only sampled once;
- (3) Choose the materials that are the nearest neighbour to the selected point.

In this work, the LHS was conducted using the design of experiment methods in the *emukit* package. The bounds of parameter space for LHS were determined based on the literature dataset of 40 ammonium salts. We chose the division level to be $m = 10$ for sampling. However, LHS samples might pick a set of descriptors that might not actually exist. We used the Nearest Neighbor method to find the closest materials (that are really available) to the LHS picks. Finally, we down-selected the 5 ammonium salts that are commercially available.

(c) Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a tool of dimensional reduction and data distribution visualization. The general idea of PCA is to find the best projection of the

original high-dimensional variables onto a limited number of principal components by orthogonal transformations. PCA was used as a dimension reduction method for visualization purpose. We can directly understand the feature similarity and distribution of our initial dataset and supplementary data selected by LHS method in a 2D scattered plot with the first two PCA components. PCA-1 and PCA-2 represent the two principal component factors with the largest and second-largest differences. The PCA was implemented using *scikit-learn* package. To be consistent with the LHS procedure, PCA transformation function was fitted with the literature dataset of 40 ammonium salts. But this PCA function was applied to both the experimental data and the PubChem dataset for visualization. The details of the PCA procedure are listed as follows:

1. Standardize the range of initial features;
2. Compute the covariance matrix to identify correlations;
3. Compute the eigenvectors and eigenvalues of the covariance matrix to identify the principal components;
4. Create a feature vector to decide which principal components to keep;
5. Recast the data along the principal components' axes."

(d) Regression models

We used the *scikit-learn* package in Random Forest (RF) regression, Gradient Boosting (GB) regression, Support Vector Machines (SVM) regression, Multilayer Perceptron (MLP) regression, and Gaussian Process (GP) regression.

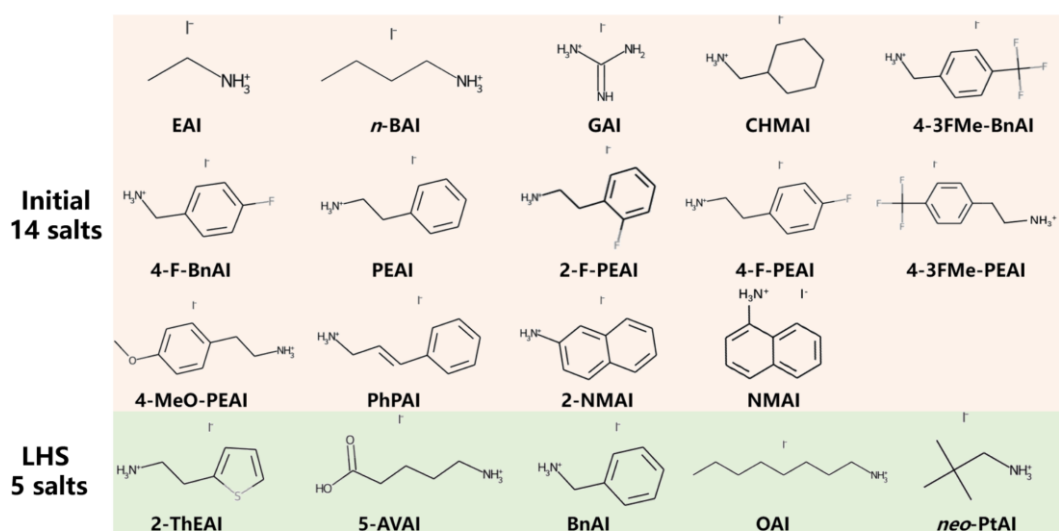
The final ensemble learning model was implemented by combining the outputs from all the regression models (i.e., the base learner) using a Random Forest regressor (i.e., the secondary learner). Secondary learning aims to learn the difference between the predicted PCEs from the base learner to the ground truth PCEs (i.e., the experiments). Via the ensemble learning model, we can usually take advantage of the regression with different architectures (e.g., tree-based, kernel-based), which reduces the possibility of overfitting in our case of a small training dataset.

3. Supplementary data and figures

3.1 Dataset for ML training

Figure S1a shows the molecular structures and abbreviations of the 14 initial and 5 LHS-selected ammonium salts. **Figure S1b** shows the formation of the 2D/3D stacked perovskite structure by ammonium salts post-treatment. We have collected a total of 46 data points with the maximum device PCE of untreated control and 19 ammonium salts treated devices with different concentrations in **Table S1**. The improvement ratio of PCE is calculated with the corresponding control device in a specific device batch to avoid batch-to-batch variation. The 46 data points were further used for ML model training.

(a)



(b)

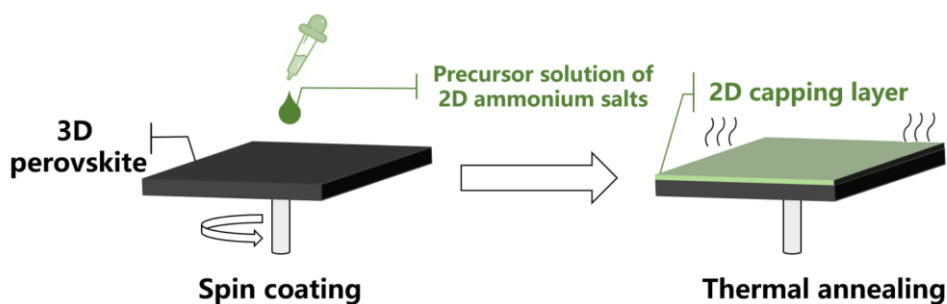


Figure S1. Acquiring of the experimental dataset for ML training. (a) The chosen ammonium salts for constructing the ML model. (b) The schematic diagram of the formation of the 2D/3D structure perovskite films.

Table S1. The statistic of PCE_{max} in 46 groups of PSCs.

	Full Name	Abbreviation	Concentration	PCE _{max} of Control device (%)	PCE _{max} of treated-devices (%)	The improvement PCE _{max} ratio in experiments
1	<i>n</i> -Butylammonium iodide	<i>n</i> -BAI	5 mg/mL (25 mM)	17.62	18.68	1.060
2	Cyclohexylmethylammonium iodide	CHMAI	5 mg/mL, (21 mM)		18.71	1.062
3	Ethylammonium iodide	EAI	5 mg/ml, (29 mM)	18.82	20.01	1.063
4			7.5 mg/mL, (43 mM)		19.94	1.060
5			10 mg/mL, (58 mM)		20.12	1.069
6			12.5 mg/mL, (72 mM)		18.30	0.972
7			15 mg/mL, (87 mM)		13.06	0.694
8	Guanidinium iodide	GAI	5 mg/mL, (27 mM)	18.82	17.75	0.943
9	4-Fluoro-phenethylammonium iodide	4-F-PEAI	5 mg/mL, (19 mM)	18.42	20.12	1.092
10	2-Fluoro-phenethylammonium iodide	2-F-PEAI	1 mg/mL, (4 mM)	20.09	18.82	0.937
11			3 mg/mL, (11 mM)		21.18	1.054
12			5 mg/mL, (19 mM)		19.76	0.984
13			7 mg/mL, (26 mM)		20.40	1.015
14			9 mg/mL, (34 mM)		20.37	1.014
15	(4-Fluorophenyl) methanaminium iodide	4-F-BnAI	5 mg/mL, (20 mM)	19.13	19.27	1.007
16			10 mg/mL, (40 mM)		20.53	1.073
17			15 mg/mL, (59 mM)		19.47	1.018
18	4-Methoxy-Phenethylammonium iodide	4-MeO-PEAI	5 mg/mL, (18 mM)	18.31	20.04	1.094
19			10 mg/mL, (36 mM)		18.73	1.023
20			15 mg/mL, (54 mM)		18.74	1.023
21	2-Phenylethan-1-aminium iodide	PEAI	3 mg/mL, (12 mM)	18.58	20.16	1.085
22			5 mg/mL, (20 mM)		19.52	1.051
23			7 mg/mL, (28 mM)		20.30	1.093
24			9 mg/mL, (36 mM)		18.56	0.999

25	4-(Trifluoromethyl)phenylmethanaminium iodide	4-3FMe-PEAI	5 mg/mL, (16 mM)	18.42	18.84	1.023
26	4-trifluoromethyl-benzenaminium iodide	4-3FMe-BnAI	5 mg/mL, (17 mM)		17.78	0.965
27	Naphthalen-2-aminium iodide	2-NMAI	5 mg/mL, (18 mM)		18.96	1.029
28	Naphthalen-1-aminium iodide	NMAI	5 mg/mL, (18 mM)		18.29	0.993
29	3-Phenylprop-2-en-1-aminium iodide	PhPAI	1 mg/mL, (4 mM)	19.94	21.77	1.092
30			3 mg/mL, (11 mM)		21.89	1.098
31			5 mg/mL, (19 mM)		20.68	1.037
32	2-(Thiophen-2-yl) ethan-1-aminium iodide	2-ThEAI	3 mg/mL, (12 mM)	19.49	19.06	0.978
33			6 mg/mL, (24 mM)		19.79	1.015
34			9 mg/mL, (35 mM)		20.36	1.045
35	5-Ammonium valeric acide iodide	5-AVAI	3 mg/mL, (12 mM)		18.45	0.947
36			6 mg/mL, (24 mM)		16.88	0.866
37			9 mg/mL, (37 mM)		14.31	0.734
38	Benzylammonium iodide	BnAI	3 mg/mL, (13 mM)		20.40	1.047
39			6 mg/mL, (26 mM)		18.57	0.953
40			9 mg/mL, (38 mM)		19.53	1.002
41	Octylammonium iodide	OAI	3 mg/mL, (12 mM)		19.61	1.006
42			6 mg/mL, (23 mM)		19.35	0.993
43			9 mg/mL, (35 mM)		19.57	1.004
44	<i>neo</i> -Pentammonium iodide	<i>neo</i> -PtAI	3 mg/mL, (14 mM)	17.92	19.70	1.099
45			6 mg/mL, (28 mM)		18.56	1.036
46			9 mg/mL, (42 mM)		18.54	1.035

3.2 Effects of concentration on PSC performance

The concentration of ammonium salt solution is an important experimental variable that affects the PCE. The best concentrations of PSC post-treatment for ethylammonium iodide (EAI), benzylammonium iodide (BnAI), phenethylammonium iodide (PEAI), and 2-fluorophenethylammonium iodide (2-F-PEAI) were 3, 9, 7, and 5 mg/mL, respectively. The $J-V$ curves of the EAI, BnAI, PEA, and 2-F-PEAI treated devices with different concentrations are presented in **Figure S2**.

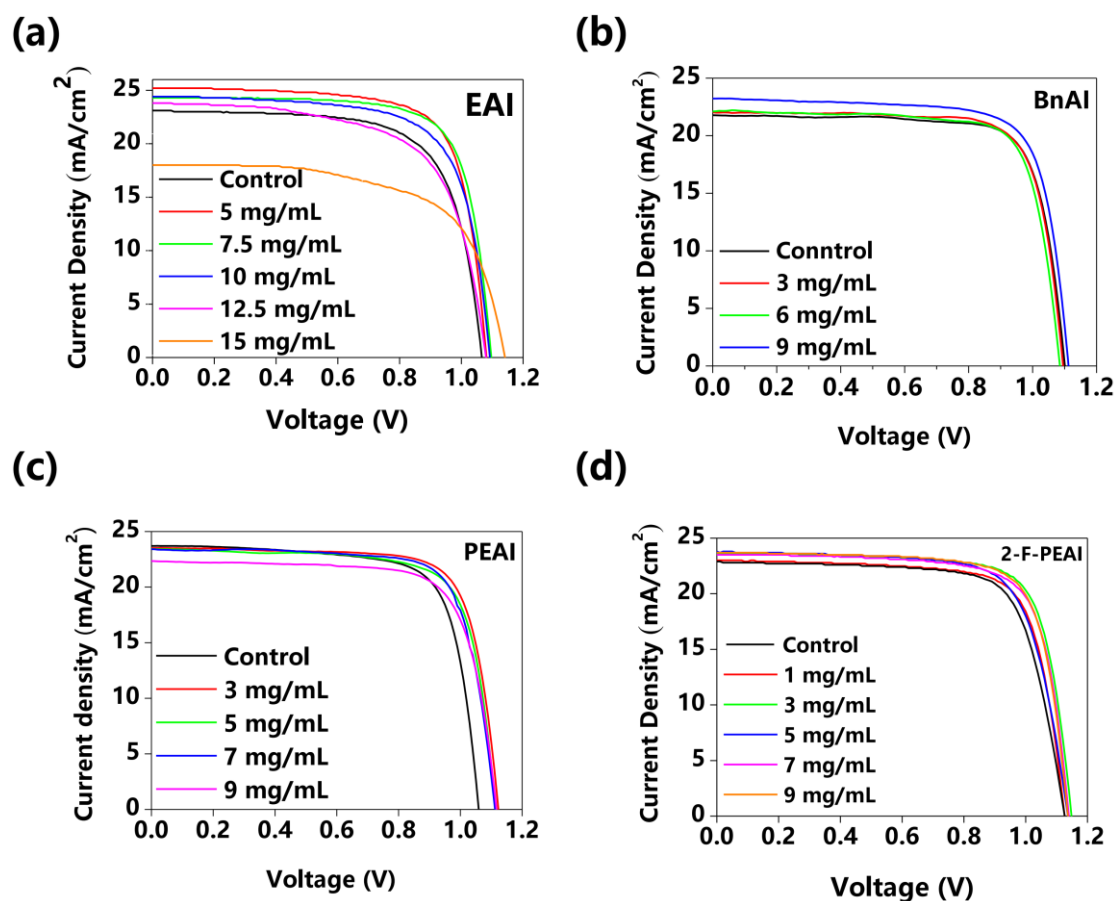


Figure S2. The $J-V$ curves of four ammonium iodide treated devices. (a) Ethylammonium iodide (EAI) treated, (b) benzylammonium iodide (BnAI) treated, (c) phenethylammonium iodide (PEAI), and (d) 2-fluorophenethylammonium iodide (2-F-PEAI) treated devices with different concentration.

3.3 Optimizations of the ML models

Our dataset has the following distinct characteristics: (a) the dataset is relatively small (a list of J - V analysis data entries for 19 materials with varying concentrations), as shown in **Table S1**, and (b) the input variables are mixed with both discrete and continuous feature descriptors (descriptors for the atom counts are discrete, and the molecular weight is continuous). The remaining five models had comparable prediction MAEs for the test dataset. To evaluate the accuracy of the model, the dataset is randomly divided into two subsets with a ratio of 4:1 for the training and the testing sets. The optimized hyperparameters of these models are shown in Table S3. To avoid overfitting, we selected the model hyperparameter using a random search with five-fold cross-validation, which is commonly used for the hyperparameter tuning of ML models. Owing to the relatively small dataset, we obtained the average and standard deviation of the predicted MAEs by running the model training procedure 200 times.

Table S2. The hyperparameters of RF, GB, SVM, MLP, GP, and Ensemble Learning regressors.

Regressor	Hyperparameters
RF	n_estimators = 20; max_depth = 5
GB	n_estimators = 20; max_depth = 4
SVM	c = 10; epsilon = 0.1
MLP	hidden_layer_size = [32, 64, 32], alpha = 0.01, activation = 'relu',
GP	ker = Matern52 (ARD = True)
Ensemble Learning	The base learners are, i.e., GB, RF, SVM, MLP, and GP regressors. The secondary learner (also called meta-learner) used is another RF model to ensemble the base learners. Thus, the outputs of the base learners are in fact the inputs of the secondary learner.

3.4 Assessments of the ML models

The MAE of the prediction results for the training set only indicates whether the ML model can fit the given dataset, whereas the MAE for the test set was a better metric used for the performance evaluation of an ML regression model. However, the MAE gap between the training and test sets should not be large. Otherwise, it would indicate the possibility of overfitting.

Table S3. The standard deviations MAE of prediction error for RF, GB, SVM, MLP, GP, and Ensemble models.

	MAE	
	Train score	Test score
RF	0.0227 ± 0.0028	0.0514 ± 0.0178
GB	0.0183 ± 0.0028	0.0507 ± 0.0174
SVM	0.0524 ± 0.0045	0.0622 ± 0.0170
MLP	0.0550 ± 0.0172	0.0650 ± 0.0238
GP	0.0242 ± 0.0057	0.0517 ± 0.0178
Ensemble	0.0116 ± 0.0020	0.0484 ± 0.0152

3.5 Model prediction for ammonium iodides in Pubchem database

To test the prediction ability of the ML ensemble model, the 112 types of ammonium iodide from Pubchem database were input into the ensembled ML model for PCE prediction. The potential of PCE improvement was attained and listed in **Table S4**. Furthermore, 4 salts from the predicted top 10 ammonium iodides and 2 salts from the bottom 10 ammonium iodide were selected to fabricate the PSCs for model validation. **Table S5** shows the feature information of the top 10 ammonium salts from the ensemble model prediction. **Table S6** shows the feature information of the bottom 10 ammonium salts from the ensemble model prediction.

Table S4. The prediction of the improvement ratio of PCE_{max} with the 112 types of ammonium iodide from the PubChem database. The salts marked with * are used in experiments for validation.

	Full Name	Predicted improvement ratio of PCE_{max}
1	2-Phenylpropan-1-aminium iodide (2-PPAI) *	1.086
2	4-Methyl-phenethylammonium iodide (4-Me-PEAI) *	1.086
3	3-Phenylpropan-1-aminium iodide (PPAI) *	1.085
4	2-(1-Cyclohexenyl) ethylammonium iodide (2-CHEAI)	1.085
5	1-Phenylethan-1-aminium iodide (1-PEAI)	1.084
6	Pentylammonium iodide (PtAI) *	1.084
7	(4-Methoxyphenyl) methanaminium iodide (4-MeO-PMAI)	1.084
8	Cyclopentanaminium iodide (CPAI)	1.084
9	Cyclohexanaminium iodide (CHAI)	1.084
10	Butan-2-aminium iodide (BtAI)	1.084
11	Butylammonium iodide	1.084
12	4,4-Difluorocyclohexan-1-aminium iodide	1.080
13	2-(1-Pyridyl) ethylammonium iodide	1.077
14	Methoxyphenethylammonium iodide	1.075
15	2-Phenylcyclopropan-1-aminium iodide	1.074
16	5-Iodopentylammonium iodide	1.074
17	(4-Vinylphenyl) methanaminium iodide	1.073
18	2-(2-Methoxyphenyl) ethan-1-aminium iodide	1.072
19	2-(3-Methoxyphenyl) ethan-1-aminium iodide	1.072
20	1-(M-tolyl) propan-2-aminium iodide	1.072
21	(3,4-Dihydronaphthalen-1-yl) methanaminium iodide	1.069
22	1-Methyl-hexylammonium iodide	1.068
23	4-Iodobutylammonium iodide	1.064
24	Heptylammonium iodide	1.060
25	6-Iodoheptylammonium iodide	1.059
26	(1,2-Dihydronaphthalen-1-yl) methanaminium iodide	1.057
27	2-Phenylcyclopent-2-en-1-aminium iodide	1.057
28	3-Iodopropylammonium iodide	1.057

29	Naphthalen-2-ylmethanaminium iodide	1.056
30	But-3-en-1-aminium iodide	1.054
31	2-(2-Allylphenyl) ethan-1-aminium iodide	1.053
32	Cyclopropanaminium iodide	1.052
33	2-(Naphthalen-2-yl) ethan-1-aminium iodide	1.052
34	2-(Naphthalen-1-yloxy) ethan-1-aminium iodide	1.051
35	1-(1-Naphthyl) ethylammonium iodide	1.050
36	But-3-yn-1-aminium iodide	1.049
37	1-(1-Naphthyl) ethylamine iodide	1.049
38	2-(Methylthio) ethan-1-aminium iodide	1.048
39	1-(4-Chlorophenyl) ethylammonium iodide	1.048
40	1-Phenylpent-4-en-2-aminium iodide	1.047
41	2,3,4,5-Tetrahydro-[1,1'-biphenyl]-4-aminium iodide	1.046
42	3,4,5,6-Tetrahydro-[1,1'-biphenyl]-3-aminium iodide	1.046
43	Anthracen-9-ylmethanaminium iodide	1.046
44	4-Chlorophenylmethylammonium iodide	1.046
45	2-(Pyren-1-yloxy) ethan-1-aminium iodide	1.045
46	2-([1,1'-Biphenyl]-4-yl) ethan-1-aminium iodide	1.045
47	2-([1,1'-Biphenyl]-3-yl) ethan-1-aminium iodide	1.045
48	2-(Perylen-3-yloxy) ethan-1-aminium iodide	1.045
49	2-Benzylcyclohexa-2,4-dien-1-aminium iodide	1.045
50	[1,1'-Biphenyl]-1(2H)-ylmethanaminium iodide	1.044
51	1-(p-tolyl) Butan-2-aminium iodide	1.044
52	6-Phenylhexan-1-aminium iodide	1.042
53	(3,4-Dihydro-[1,1'-biphenyl]-2-yl) methanaminium iodide	1.042
54	Decylammonium iodide	1.042
55	4-Bromophenylmethylammonium iodide	1.041
56	4-Iodophenylmethylammonium iodide	1.041
57	Nonylammonium iodide	1.041
58	2-([2,2'-Bithiophen]-5-ylthio) ethan-1-aminium iodide	1.040
59	2-Ethyl-hexylammonium iodide	1.039
60	2-(3-Fluorophenyl) ethan-1-aminium iodide	1.039
61	2,2-Diphenylethan-1-aminium iodide	1.038
62	Pyrene-butylammonium iodide	1.038
63	3-(Naphthalen-1-yloxy) propan-1-aminium iodide	1.037
64	2-Phenylcycloheptan-1-aminium iodide	1.037
65	3-([2,2-Bithiophen]-4-yl) propan-1-aminium iodide	1.036
66	3-(Aminomethyl)pyridinium iodide	1.036
67	4-(Aminomethyl)pyridinium iodide	1.036
68	4-Methyl-1-phenylpent-4-en-2-aminium iodide	1.036
69	2-Cyclopentyl-2-phenylethan-1-aminium iodide	1.034
70	Pyridine-3-ylmethanaminium iodide	1.034
71	Pyridine-2-ylmethanaminium iodide	1.034
72	4-(9H-fluoren-9-yl) butan-1-aminium iodide	1.034
73	3-(Pyren-1-yloxy) propan-1-aminium iodide	1.033
74	2-(4-Benzylphenyl) ethan-1-aminium iodide	1.033
75	2-Cyclohexyl-2-phenylethan-1-aminium iodide	1.033
76	2,3-Diphenylpropan-1-aminium iodide	1.033

77	4-Phenylhex-5-en-1-aminium iodide	1.027
78	4-(Pyren-1-yloxy) butan-1-aminium iodide	1.026
79	2-Bromoethylammonium iodide	1.025
80	1-Cyclohexyl-3-phenylpropan-2-aminium iodide	1.025
81	2-Iodoethylammonium iodide	1.023
82	3-Ethyl-3-phenylpentan-1-aminium iodide	1.022
83	2-(2-(Trifluoromethyl) phenyl) ethan-1-aminium iodide	1.019
84	2-(3-(Trifluoromethyl) phenyl) ethan-1-aminium iodide	1.019
85	2-(4-(Trifluoromethyl) phenyl) ethan-1-aminium iodide	1.019
86	2-(3,5-Difluorophenyl) ethan-1-aminium iodide	1.018
87	2,2-Difluoroethan-1-aminium iodide	1.017
88	2,2,2-Trifluoroethan-1-aminium iodide	1.014
89	1-Methyl-1H-imidazole-3-ium iodide	1.013
90	2-(2,4,6-Trifluorophenyl) ethan-1-aminium iodide	1.013
91	2,2,3,3,3-Pentafluoropropan-1-aminium iodide	1.008
92	4,4'-Bipyridine iodide	1.007
93	Benzimidazolium iodide	1.005
94	3-((1,1,1,3,3,3-Hexafluoro-2-(trifluoromethyl) propan-2-yl) oxy) propan-1-aminium iodide	0.992
95	2,2'-Bithiophene iodide	0.984
96	4-(Aminomethyl)piperidinium iodide	0.983
97	3-(Aminomethyl)piperidinium iodide	0.983
98	Piperazinium iodide	0.974
99	3-Aminopyrrolidinium iodide	0.974
100	2-Hydroxyethan-1-aminium iodide	0.973
101	2-(4-Nitrophenyl) ethan-1-aminium iodide	0.967
102	Methylhydrazinium iodide	0.961
103	Formamidine iodide (FAI)	0.961
104	1-Aminoethan-1-iminium iodide (1-AEI) *	0.959
105	Aminoiodomethaniminium iodide (AMAI)	0.959
106	4-Carboxy-cyclohexyl-ethylammonium iodide (4-CCHEAI)	0.954
107	cis-4-Aminocyclohexanecarboxylic acid iodide (CACHCAI)	0.952
108	trans-4-Aminocyclohexanecarboxylic acid iodide (TACHCAI)	0.952
109	(2-Carboxy) ethylammonium iodide (2-CEAI) *	0.947
110	(2-Carboxy) butylammonium iodide (2-CBAI)	0.946
111	(2-Carboxy) propylammonium iodide (2-CPAI)	0.946
112	3-Carboxypropan-1-aminium iodide (3-CPAI)	0.940

Table S5. Predicted top-10 ammonium iodides.

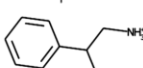
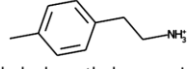
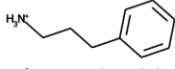
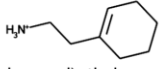
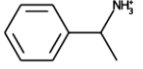
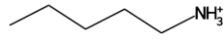
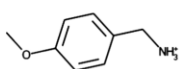
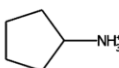
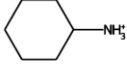
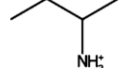
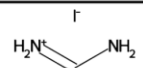
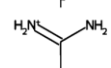
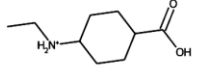
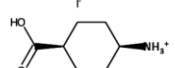
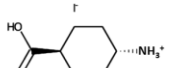
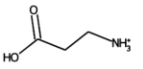
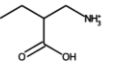
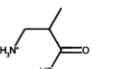
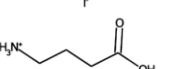
Ammonium salts	H bond donor	H atom	MolLogP	Complexity	ratio
 2-Phenylpropan-1-aminium iodide	1	14	-2.27	186.43	1.0860 ± 0.0500
 4-Methyl-phenethylammonium iodide	1	14	-2.22	193.22	1.0858 ± 0.0500
 3-Phenylpropan-1-aminium iodide	1	14	-2.13	174.18	1.0851 ± 0.0500
 2-(1-Cyclohexenyl)ethylammonium iodide	1	16	-1.88	109.83	1.0850 ± 0.0500
 1-Phenylethan-1-aminium iodide	1	12	-2.00	171.87	1.0838 ± 0.0500
 Pentylammonium iodide	1	14	-2.58	20.04	1.0838 ± 0.0500
 (4-Methoxyphenyl)methanaminium iodide	1	12	-2.55	174.75	1.0838 ± 0.0500
 Cyclopentanaminium iodide	1	12	-2.83	41.34	1.0838 ± 0.0500
 Cyclohexanaminium iodide	1	14	-2.44	50.49	1.0837 ± 0.0500
 Butan-2-aminium iodide	1	12	-2.97	21.51	1.0837 ± 0.0500

Table S6. Predicted bottom-10 ammonium iodides.

Ammonium salts	H bond donor	H atom	MolLogP	Complexity	ratio
 Formamidine iodide	2	5	-5.26	13.51	0.9613 ± 0.0500
 1-Aminoethan-1-iminium iodide	2	7	-4.87	30.63	0.9599 ± 0.0500
 Aminoiodomethaniminium iodide	2	4	-4.50	30.63	0.9598 ± 0.0500
 4-Carboxy-cyclohexyl-ethylammonium iodide	2	18	-2.78	155.69	0.9544 ± 0.0500
 cis-4-Aminocyclohexanecarboxylic acid iodide	2	13	-2.41	132.40	0.9524 ± 0.0500
 trans-4-Aminocyclohexanecarboxylic acid iodide	2	13	-2.41	132.40	0.9524 ± 0.0500
 2-Carboxyethylammonium iodide	2	8	-4.29	56.89	0.9470 ± 0.0500
 2-Carboxybutylammonium iodide	2	12	-3.65	83.03	0.9467 ± 0.0500
 2-Carboxypropylammonium iodide	2	10	-4.04	76.44	0.9462 ± 0.0500
 3-Carboxypropan-1-aminium iodide	2	10	-3.90	67.06	0.9391 ± 0.0500

3.6 Model validation in FAMACs-based PSCs

Figure S3 shows the statistical PCE of FAMACs-based PSCs with and without ammonium treatments. The optimal concentrations of 2-Phenylpropan-1-aminium iodide (2-PPAI), 2-(p-tolyl) ethan-1-aminium iodide (4-Me-PEAI), phenylpropan-1-aminium iodide (PPAI) and pentylammonium iodide (PtAI) were 1 mg/mL, 3 mg/mL, 7 mg/mL, and 1 mg/mL, respectively. The two “bad” salts, 1-Aminoethan-1-iminium iodide (1-AI) and (2-carboxy) ethylammonium iodide (2-CEAI), both show negative effects on PCE even at a low concentration of 1 mg/mL. The results of experimental validation are consistent with the prediction of our ensemble ML model.

Table S7 shows the average and the best $J-V$ parameters of the ammonium-treated devices. the improvement ratio of PCE was calculated in each group. The best device was obtained by the 2-PPAI treatment, which achieved a PCE of 22.36% with a PCE improvement ratio of 1.15.

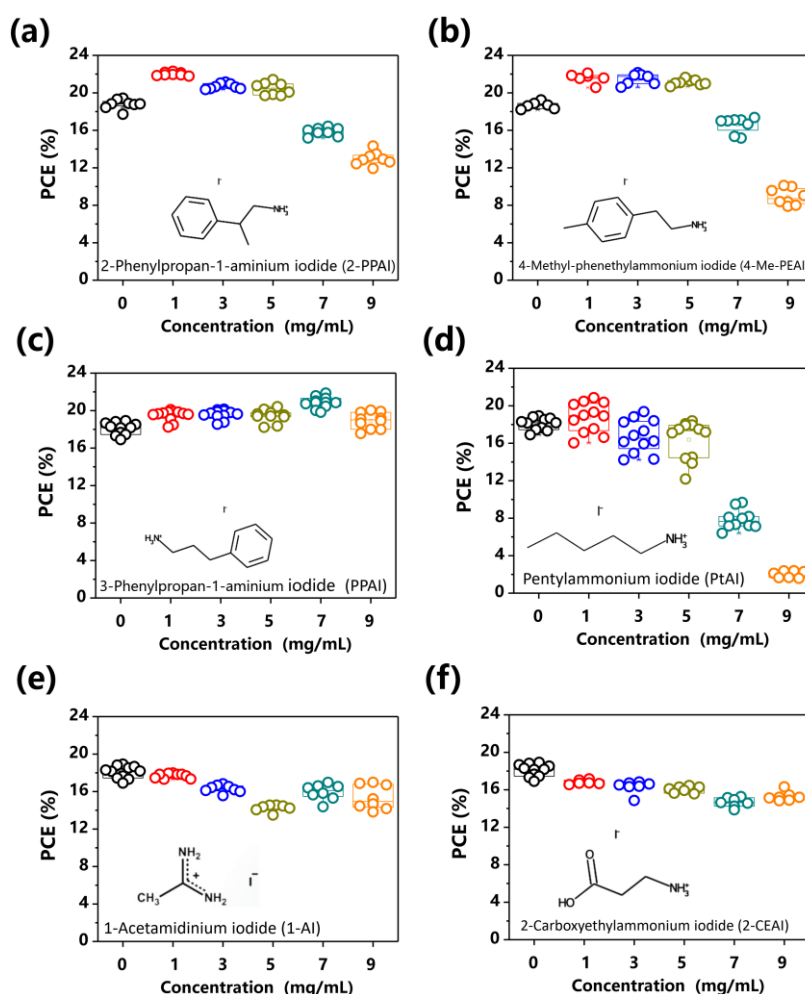
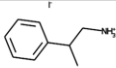
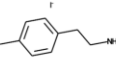
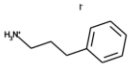
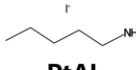
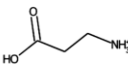


Figure S3. The statistical analysis of $J-V$ curve data of FAMACs-based control, (a) 2-PPAI-treated (b) 4-Me-PEAI-treated, (c) PPAI-treated, (d) PtAI, (e) 1-AI, and (f) 2-CEAI-treated PSCs.

Table S7. Photovoltaic performance of the FAMACs-based PSCs treated with different ammonium salts.

		V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)	Improvement Ratio
	best	1.16	24.22	79.57	22.36	1.15
2-PPAI	Average	1.15±0.01	24.06±0.16	79.04±0.56	22.00±0.36	
	best	1.17	23.84	79.49	22.17	1.15
4-Me-PEAI	Average	1.15±0.02	23.55±0.29	79.28±0.19	21.48±0.69	
	best	1.15	24.30	78.38	21.87	1.13
PPAI	Average	1.14±0.01	23.76±0.65	76.80±1.58	20.86±1.01	
	best	1.13	24.45	75.61	20.89	1.08
PtAI	Average	1.12±0.01	23.52±0.93	71.05±4.56	18.74±2.15	
	best	1.06	23.44	72.46	18.00	0.93
1-AI	Average	1.03±0.03	23.27±0.17	73.67±1.51	17.69±0.31	
	best	1.07	22.85	70.40	17.16	0.88
2-CEAI	Average	1.04±0.03	22.75±0.10	70.79±2.66	16.81±0.35	

3.7 Model validation in FAMA-based PSCs

Although the ML model was trained using data of FAMACs-based PSCs, the prediction can be extended to other perovskite composition. **Figure S4** shows the best J - V curve and the max power tracking of FAMA-based control and treated PSC devices using 2-PPAI, 4-Me-PEAI, and PPAI. The best device was obtained by the 2-PPAI treatment, which achieved a PCE of 24.47% and a stable power output of 24.17% at 1.00 V. **Table S8** shows the average and the best photovoltaic performance parameters of the FAMA-based devices with 2-PPAI, 4-Me-PEAI, and PPAI treatment. These results show that the prediction of the ensemble model can be utilized as a reference for improving the performance of other PSCs.

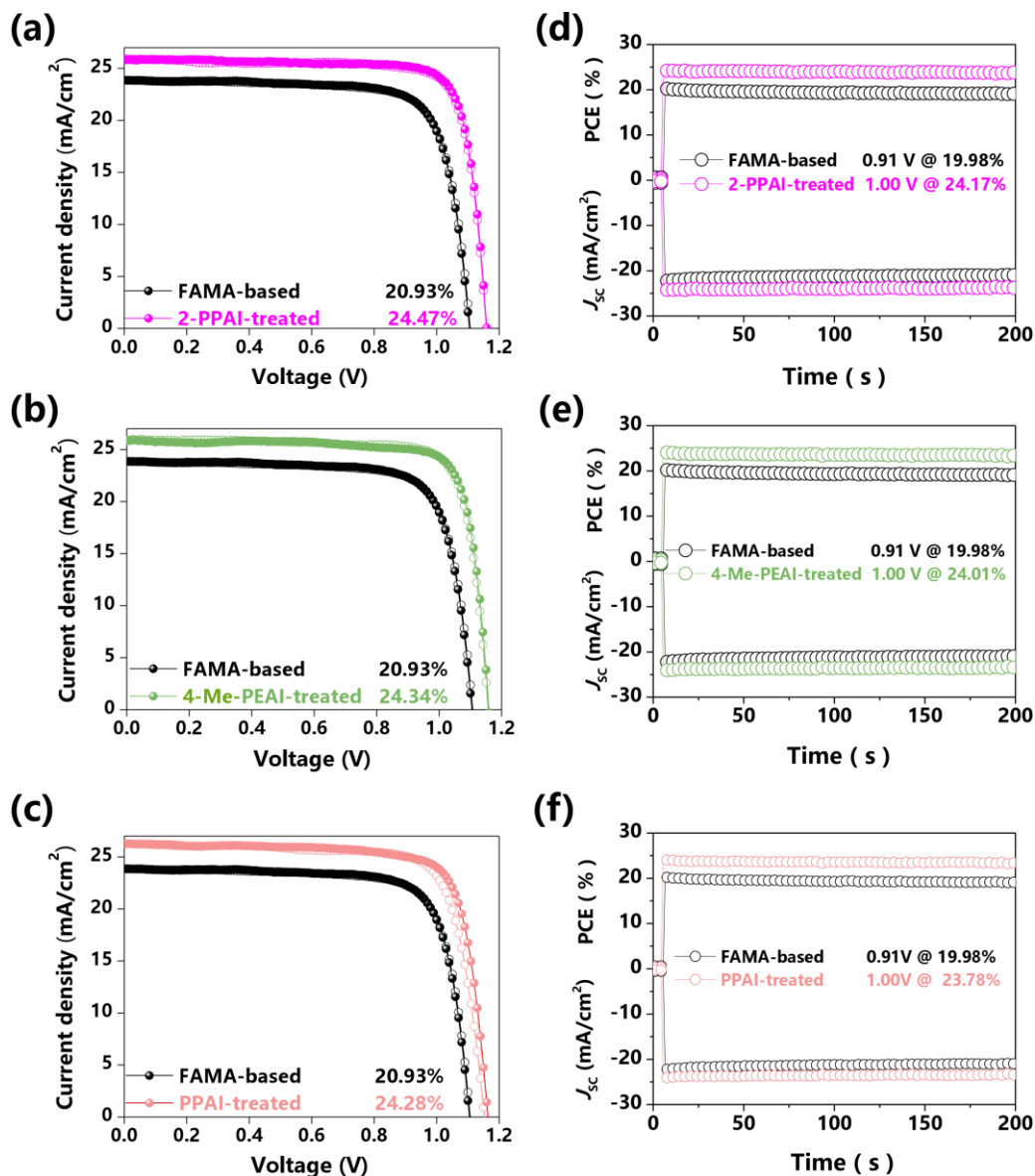
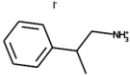
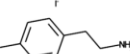
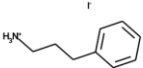


Figure S4. J - V curves and stabilized power output of FAMA-based PSCs with 2-PPAI, 4-Me-PEAI, and PPAI treatments. (a) J - V curves. (b) stabilized power output trackings.

Table S8. Photovoltaic performance of the FAMA-based PSCs with 2-PPAI, 4-Me-PEAI, and PPAI treatments.

		V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)
FAMA-based control	best	1.12	24.43	76.19	20.93
	Average	1.12±0.02	24.12±0.45	74.86±2.13	20.23±0.70
 2-PPAI	best	1.16	25.92	81.39	24.47
	Average	1.15±0.01	25.49±0.48	80.30±1.09	23.68±0.79
 4-Me-PEAI	best	1.16	25.88	81.08	24.34
	Average	1.15±0.02	25.30±0.76	79.66±1.42	23.42±0.92
 PPAI	best	1.17	25.82	80.41	24.29
	Average	1.15±0.02	25.25±0.57	78.20±2.21	22.87±1.42